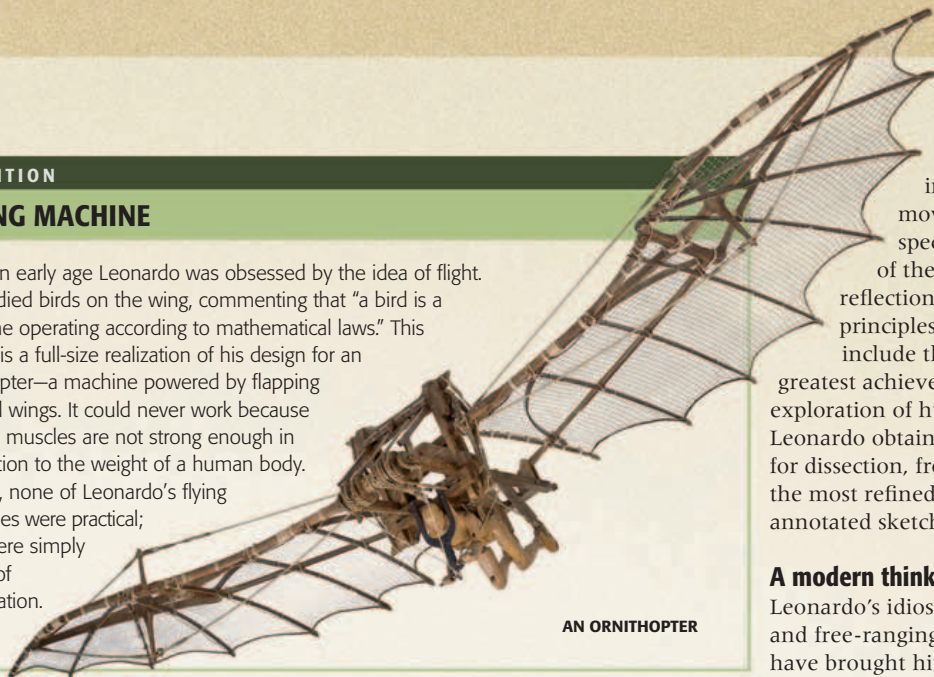


INVENTION

FLYING MACHINE

From an early age Leonardo was obsessed by the idea of flight. He studied birds on the wing, commenting that “a bird is a machine operating according to mathematical laws.” This model is a full-size realization of his design for an ornithopter—a machine powered by flapping artificial wings. It could never work because human muscles are not strong enough in proportion to the weight of a human body. Indeed, none of Leonardo’s flying machines were practical; they were simply flights of imagination.



AN ORNITHOPTER

experiments. They include studies of the movement of water and speculations on the nature of the cosmos, as well as reflections on fossils, and on the principles of flight. They also include the record of one of his greatest achievements, the systematic exploration of human anatomy. Leonardo obtained about 30 corpses for dissection, from which he produced the most refined, accurate, and annotated sketches.

A modern thinker

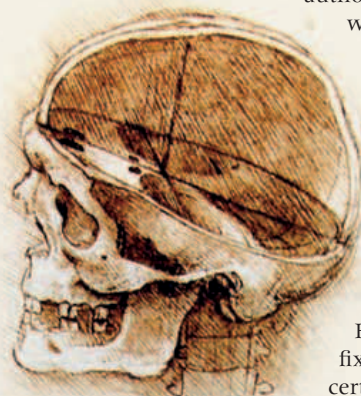
Leonardo’s idiosyncratic personality and free-ranging mind could easily have brought him into conflict with the authorities. His homosexuality

was both a sin and a crime in the society he inhabited, even if it was not generally disapproved of in sophisticated circles. Dissecting corpses was of dubious legality and almost got him into trouble in Rome in about 1515. He seems to have had no fixed religious beliefs and certainly did not allow

Christian doctrine to set limits to his thinking. In his declining years, Leonardo was respected, even revered. Around 1515 he appears to have suffered a stroke. Unable to undertake major works, he found a sympathetic patron in the French King Francis I, who appreciated the prestige of having such a renowned figure in his employ. It is reported that Francis said he could never believe there was another man born in this world who knew as much as Leonardo. Few of Leonardo’s paintings have survived, but their scarcity has if anything enhanced his legendary status.

Anatomical drawing

Leonardo’s sketch of a cross section of a human skull showed impressive anatomical details, which are due to the autopsies he preformed.



Milan. Judging that the warlike Milanese ruler might have more need of a military engineer than of an artist, Leonardo listed at length the “infinite variety of machines for attack and defense” that he could create, including armored cars, portable bridges, and cannons, before mentioning as an afterthought his skills at painting and sculpture.

A man of talent

Leonardo’s all-around skills, including his ingenuity at designing machines, were exactly what Renaissance princes were looking for, and he rarely lacked employment for the rest of his life. Even at the end of his career, he was engaged by the French monarchy as firstly “engineer” and then “painter and architect.” Leonardo’s boundless imagination and experimentation led to many failures and disappointments. For Ludovico Sforza he designed one of the largest equestrian statues ever conceived, but it is unlikely that it could have been cast and only a clay model was built. His *Last Supper*, considered one of the ultimate masterpieces of Western art, was painted using an innovative technique that meant it started decaying almost as soon as it was finished. In the service of the ruler of Florence from 1503, he produced a

grandiose project for a canal linking the city to the sea, which was never built, and he also undertook to paint the Anghiari fresco for the Palazzo Vecchio, which was never completed. The machines he dreamed of were ingenious and forward-thinking but mostly impractical.

In fact, much of Leonardo’s most remarkable work consisted of writings and sketches in his private notebooks, material that remained unpublished until long after his death. It was in these notebooks that Leonardo’s fertile, inquiring, adventurous mind and eye found full expression. The pages are packed with an astonishing number of ideas, observations, projects, and

“Art is **never finished**, only abandoned.”

LEONARDO DA VINCI



The Last Supper

Leonardo painted *The Last Supper* in the refectory of the church of Saint Maria delle Grazie in Milan, using a mix of tempera and oil, rather than a traditional fresco technique. As a result, the paint soon began to flake off.

TIMELINE

- **April 15, 1452** Born near Vinci, a small town outside Florence, the illegitimate son of a peasant woman and a local lawyer.
- **1466** Apprenticed to Andrea del Verrocchio.
- **April 1476** Accused of homosexual activity, an offense that could carry the death penalty.
- **1477** Sets up his own studio in Florence.
- **1481** Commissioned to paint the *Adoration of the Magi* for the monastery of San Donato.
- **1482** Enters the service of Ludovico Sforza, known as Ludovico il Moro, the ruler of Milan.
- **April 25, 1483** Commissioned to paint an altarpiece for the church of San Francesco in Milan, now known as the *Virgin of the Rocks*.
- **1487** Starts to write a notebook, which he fills with sketches and his imaginative thoughts.
- **1490** Engages Giacomo Caprotti, known as Salai, as a young assistant who almost certainly becomes his long-term lover.
- **1491** Paints a portrait of Ludovico’s mistress, now known as *The Lady with the Ermine*.
- **1492** Produces the famous drawing of the Vitruvian Man, showing the idealized proportions of the human body.
- **1498** Completes fresco of *The Last Supper*.
- **1502** Enters the service of the lord of the Romagna, Cesare Borgia, as a military architect and engineer.
- **1503–04** Returns to Florence and paints the *Mona Lisa*, probably a portrait of the wife of silk merchant Francesco del Giocondo. Works on a failed project for a canal to give Florence direct access to the sea.
- **1504** Starts work on a fresco, the *Battle of Anghiari*, for the Council Hall in the Palazzo Vecchio, Florence. It is never completed.
- **July 1504** His father dies. Family intrigue denies Leonardo a share of the inheritance.
- **1506** Returns to Milan and enters the pay of Louis XII of France.
- **1508** Produces obsessive images in his notebooks of whirlwinds and swirling rainstorms.
- **1509** The mathematician Pacioli’s *De Divina Proportione* is published with geometrical illustrations by Leonardo.
- **1510** Completes *The Virgin and Child with St. Anne*.
- **1513** Moves to Rome, entering the service of Giuliano de’ Medici; settles in the Villa Belvedere, lent to him by Giuliano’s brother, Pope Leo X.
- **1515** Suffers an illness, possibly a stroke, that leaves him paralyzed down one side.
- **1516** Appointed Premier Engineer, Painter, and Architect to Francis I of France. Lives in Clos Luce near the royal chateau at Amboise.
- **May 2, 1519** Dies at Amboise.



LA GIOCONDA